

Backend Installation & Setup Guide

Automated Interview Question Generator



Prerequisites

Before you begin, ensure you have the following installed on your system:

- Windows 10 or Windows 11
- Python 3.10 or higher (with pip)

You can verify your Python installation by opening Command Prompt and running:

```
python --version
```



First-Time Setup

Follow these steps carefully to set up the backend server for the first time.

Step 1: Navigate to the Backend Folder

- Open **File Explorer** and navigate to your project directory
- Locate the **backend** folder
- Click on the address bar at the top of File Explorer
- Type `cmd` and press **Enter**

This will open a Command Prompt window directly in the backend directory.

Step 2: Create and Activate Virtual Environment

A virtual environment keeps your project dependencies isolated from other Python projects.

In the Command Prompt window that just opened, run:

```
python -m venv venv
```

Wait for the environment to be created (this may take 30-60 seconds).

Once created, activate it by running:

```
venv\Scripts\activate
```

✅ **Success indicator:** Your command prompt should now show **(venv)** at the beginning of the line.

Step 3: Install Dependencies

With the virtual environment activated, install all required packages:

```
pip install --upgrade pip
```

```
pip install -r requirements.txt
```

This will install:

project dependencies listed in requirements.txt

🕒 **Time required:** 2-5 minutes depending on your internet speed.

Step 4: Configure Groq API Key

The backend uses Groq AI to generate interview questions. You need to configure your API key securely.

Add Your API Key

Add the following line to the `.env` file:

```
GROQ_API_KEY= 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
```

Step 5: Run the FastAPI Server

Now you're ready to start the backend server.

In the same Command Prompt window (with venv activated), run:

uvicorn main:app --reload

Command breakdown:

- `main:app` – Tells uvicorn to look for the `app` object in `main.py`
- `--reload` – Automatically restarts the server when code changes are detected (useful for development)

✅ **Success indicator:** You should see output similar to:

INFO: Uvicorn running on <http://127.0.0.1:8000> (Press CTRL+C to quit)
INFO: Started reloader process
INFO: Started server process
INFO: Waiting for application startup.
INFO: Application startup complete.

Step 6: Test the API

Open your web browser and navigate to:

API Documentation (Swagger UI):

<http://127.0.0.1:8000/docs>

This interactive documentation allows you to:

- View all available API endpoints
- Test endpoints directly from the browser
- See request/response schemas
- Validate that Groq integration is working correctly

Root endpoint:

<http://127.0.0.1:8000/>

You should see a response confirming the API is running.

Running the Server After First-Time Setup

Once you've completed the first-time setup, subsequent runs are much simpler.

Quick Start Steps

1. Open Command Prompt in the **backend** folder
 - Navigate to the folder in File Explorer
 - Click the address bar, type `cmd`, press Enter
2. Activate the virtual environment:

`venv\Scripts\activate`

3. Start the server:

`uvicorn main:app --reload`

4. Open <http://127.0.0.1:8000/docs> in your browser

 **Time required:** Less than 30 seconds

✂ Troubleshooting

Issue 1: "python is not recognized as an internal or external command"

Solution:

- Python is not installed or not added to PATH
- Download Python from <https://python.org>
- During installation, check "Add Python to PATH"
- Restart Command Prompt after installation

Issue 2: "`venv\ Scripts\ activate` is not recognized"

Solution:

- Make sure you're in the correct directory (backend folder)
- Verify the `venv` folder exists
- If not, recreate it: `python -m venv venv`

Issue 3: "ModuleNotFoundError: No module named 'fastapi'"

Solution:

- Virtual environment is not activated
- Run `venv\textbackslash Scripts\textbackslash activate first`
- Then reinstall requirements: `pip install -r requirements.txt`

Issue 4: "groq.GroqError: The api_key client option must be set"

Solution:

- `.env` file is missing or in the wrong location
- API key is not set correctly in `.env`
- Ensure the line is: `GROQ_API_KEY=your_key_here` (no spaces, no quotes)
- Restart the server after editing `.env`

Issue 5: "Port 8000 is already in use"

Solution:

- Another process is using port 8000
- Option 1: Stop the other process
- Option 2: Use a different port: `uvicorn main:app --reload --port 8001`

Additional Notes

Stopping the Server

To stop the running server:

- Press **CTRL + C** in the Command Prompt window
- Wait for the server to shut down gracefully

Deactivating Virtual Environment

Updating Dependencies

If `requirements.txt` changes:

1. Activate virtual environment:

```
venv\textbackslash Scripts\textbackslash activate
```

2. Update packages:

```
pip install -r requirements.txt --upgrade
```

Production Deployment

For production environments:

- Remove `--reload` flag: `uvicorn main:app --host 0.0.0.0 --port 8000`
- Use environment variables instead of `.env` file
- Consider using process managers like PM2 or systemd
- Set up reverse proxy with Nginx
- Enable HTTPS/TLS certificates



Support

If you encounter issues not covered in this guide:

Email: premlawrence54@gmail.com

Phone: +91 7483135876